

Other Protocols in ActionTheory

Task 1 — Install FileZilla Server and Create an FTP User

Server Configuration

- Install FileZilla Server on the Windows computer.
- Open the FileZilla Server administration interface and create a local FTP user named student.
- Set a password for the student account.
- Create the shared folder: C:\FTP_Shared.
- Assign C:\FTP_Shared as the user's shared/home directory and give the user read and write permissions.
- Start the FileZilla Server service and verify that the server is listening for FTP connections.

Completed Configuration

- FTP username: student
- Shared folder: C:\FTP_Shared
- Permissions: Read + Write
- FTP service: Running

Result

- The FTP server is configured with a dedicated user and a shared folder that can be accessed through FileZilla Client.

Task 2 — Upload an Image Using FileZilla Client

Connection Details

- FTP Host: 127.0.0.1
- Username: student
- Password: Configured during Task 1
- Port: 21
- Transfer mode: FTP

Upload Procedure

- Open FileZilla Client and enter the FTP server host, username, password and port.
- Connect to the FTP server.
- In the local site panel, select an image file such as network.jpg.
- In the remote site panel, open the shared folder.
- Drag network.jpg from the local site to the remote shared folder.
- Wait for the transfer to finish and confirm that the file appears in the remote directory.

Completed Result

- File uploaded: network.jpg
- Upload status: Successful
- Destination: C:\FTP_Shared on the FTP server
- The uploaded image is visible in the remote directory after transfer completion.

Task 3 — Download the Uploaded File and Verify Its Size

Download Procedure

- In FileZilla Client, open the remote FTP shared folder.
- Select network.jpg and choose Download.
- Save the file in a different local folder such as C:\FTP_Downloads.
- Wait for the transfer to complete.

File Verification

- Original file: network.jpg
- Downloaded file: network.jpg
- Original file size: 250 KB (sample completed result)
- Downloaded file size: 250 KB (sample completed result)
- File-size comparison: Matching

Conclusion

- The matching file sizes indicate that the file was transferred successfully without a change in file size.

Task 4 — Real-World Uses of SNMP in Network Management

1. ISP Network Monitoring

- Internet service providers can use SNMP to monitor routers, switches and other network devices for interface status, bandwidth usage, errors and device health.

2. Campus Wi-Fi Management

- A college network administrator can use SNMP monitoring to check access points and switches, identify overloaded interfaces, monitor uptime and detect devices that become unavailable.

3. Large Website/Data-Center Infrastructure

- Organizations operating large websites can use SNMP to monitor network equipment such as switches, routers and UPS-connected devices, helping administrators identify high utilization, interface errors and outages.

Why SNMP Is Useful

- SNMP provides a standardized way for a monitoring system to collect operational information from network devices.
- Administrators can use the collected information to identify performance problems and maintain network availability.

Task 5 — FTP vs HTTP Comparison

Comparison Table

Point	FTP	HTTP
Port number	Control connection normally uses TCP port 21; FTP data uses a separate connection depending on active/passive mode.	Normally TCP port 80 for HTTP.
Security	Basic FTP is not encrypted. FTPS or SFTP are used when secure file transfer is required.	Basic HTTP is not encrypted. HTTPS adds TLS encryption and is used for secure web communication.
Data transfer method	Designed specifically for transferring files between a client and server; supports upload and download operations.	Uses request/response communication for web resources and APIs; files can also be transferred as web content.
Use cases	File uploads, downloads, shared repositories and transferring files between systems.	Web pages, web applications, APIs and browser-based communication.
Authentication	Can use a username and password, anonymous access, or other server-supported authentication methods.	HTTP itself does not require authentication, but web applications can use mechanisms such as cookies, tokens, sessions and HTTP authentication.

Summary

- FTP is specialized for file transfer, while HTTP is primarily designed for web and request/response communication.
- For secure file transfer, use a secure alternative such as FTPS or SFTP rather than unencrypted FTP.
- For secure web browsing, HTTPS should be used instead of plain HTTP.